

Supplementary Information

Learning which edges to remove under a budget:
Counterfactual inverse link prediction for utility-preserving
social-network sparsification

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1 S1 Dataset acquisition and preprocessing

SNAP (LastFM, GitHub) and MUSAE (Facebook) public releases [Rozemberczki et al., 2021]. Symmetrization; self-loop and duplicate undirected-edge removal; multi-hot bag-of-words features.

2 S2 Dataset audit and statistics

Statistic	LastFM	Facebook	GitHub
Nodes	7624	22470	37700
Raw edge records	55612	341646	578006
Unique undirected edges	27806	170823	289003
Self-loops in tensor	0	0	0
Average degree	7.294	15.205	15.332
Median degree	4.0	7.0	6.0
Maximum degree	216	709	9458
Connected components	1	1	1
Giant-component ratio	1.000	1.000	1.000
Avg. clustering	0.219	0.360	0.168
Classes	18	4	2
Minority-class proportion	0.0021	0.1481	0.2583
Feature dimension	7842	4714	4005
Feature density	0.0504	0.0030	0.0046
Label homophily	0.874	0.885	0.845
Graph density	9.57e-04	6.77e-04	4.07e-04

Processed graphs contain no self-loops. Directed tensor encodings store both directions of each undirected edge; the unique undirected count is the scientific edge set used for budgets.

3 S3 Train/development/test splits

Seeds 0–9; stratified 60/20/20 train/development/test; masks under `data/splits/`. The middle partition is termed the development set throughout.

4 S4 Leakage audit

Stage	Train	Dev	Test	Permitted use
Features	Yes	Yes	Yes	Shared X ; no label-dependent featurization
Splits	Yes	Yes	Yes	Stratify masks only
Teacher Δ_{task}	Yes	Yes (TD mask)	No	CE on $\text{train} \cup \text{development}$
Scorer fit	Yes	Used in grid	No	Fixed epoch loops in reported grid
Teacher size n	—	Yes (Facebook)	No	Select on development Macro-F1 @50%
Final eval	—	—	Yes	After configuration is fixed

Note. The middle partition is a development set, not an independent validation hold-out: it participates in teacher-target construction and Facebook teacher-size selection. This is not test leakage, but the protocol is not fully validation-isolated. A stricter design would construct teacher targets from training labels only and reserve development exclusively for configuration selection.

5 S5 Counterfactual teacher

CILP uses a composite multi-criteria teacher that combines deletion-induced effects with structural proxies. `ExactCounterfactualTeacher.score_edges` samples n edges and returns y^{cf} plus component vectors. The task-only teacher uses only the classification-loss component under the same encoder and concatenation fusion.

6 S6 Exact deletion-importance criteria

Task loss (deletion-induced). With frozen encoder/classifier, $\Delta_{\text{task}} = \text{CE}_{TD}(G_{-e}) - \text{CE}_{TD}(G)$ on $\text{train} \cup \text{development}$ nodes.

Connectivity change (deletion-induced).

$$\Delta_{\text{conn}}(e=\{u, v\}) = \mathbf{1}[e \text{ is a bridge}] + (\# \text{comp}(G_{-e}) - \# \text{comp}(G)) + \max(0, \text{giant}(G) - \text{giant}(G_{-e}))/|V|.$$

Representation shift (deletion-induced). $\Delta_{\text{repr}}(e) = \text{mean}_i \|h_i(G) - h_i(G_{-e})\|_2$.

Group impact (deletion-induced). $\Delta_{\text{group}}(e) = \max(0, F1_{\text{dev}}^{\text{worst}}(G) - F1_{\text{dev}}^{\text{worst}}(G_{-e}))$.

Community-boundary proxy (structural proxy). Implemented as `community_effect` in `src/counterfactual/exact_teacher.py`. A Louvain partition C is computed *once* on the full undirected graph G via `community.best_partition` (python-louvain) and reused for all sampled edges; the partition is not recomputed after each deletion. Let $m = |E|$, $d_u = \deg(u)$, $d_v = \deg(v)$, $a_{uv} = 1$, and $\text{same} = \mathbf{1}[C(u) = C(v)]$. Define

$$\text{expected} = \frac{d_u d_v}{2m}, \quad \text{local} = \begin{cases} a_{uv} - \text{expected} & \text{if same} = 1, \\ |a_{uv} - \text{expected}| & \text{if same} = 0, \end{cases} \quad \text{inter} = \begin{cases} 0 & \text{if same} = 1, \\ 1 & \text{if same} = 0. \end{cases}$$

The returned score is

$$\Delta_{\text{comm}}(e) = \max(0, \text{local}) + \text{inter}.$$

Higher values indicate greater estimated removal harm. The explicit inter term assigns additional harm to *inter-community* edges, so the criterion preferentially protects community-boundary connectors rather than maximizing modularity after sparsification. We therefore call it a community-boundary proxy, not a direct community-preservation metric. The modularity value $Q(C, G)$ is computed in code but is not used in the returned score. If python-louvain is unavailable, the fallback is $\Delta_{\text{comm}} = |\text{clust}(u; G) - \text{clust}(u; G_{-e})| + |\text{clust}(v; G) - \text{clust}(v; G_{-e})|$. Reported experiments used the Louvain path.

Degree-based spectral proxy (structural proxy). For all currently reported datasets (LastFM, Facebook, and GitHub), $|V| > 400$, so

$$\Delta_{\text{spec}}(e=\{u, v\}) = \frac{1}{\max(d_u, 1)} + \frac{1}{\max(d_v, 1)}.$$

This is a degree-based spectral surrogate, not a recomputation of Laplacian eigenvalue change. The eigenspectrum variant $\|\lambda_{\text{norm}}[1:4](G) - \lambda_{\text{norm}}[1:4](G_{-e})\|_2$ is used only when $|V| \leq 400$ and was not used for the three reported datasets. Function: `spectral_effect` in the same module.

Aggregation.

$$y_e^{\text{cf}} = \minmax\left(\sum_k \beta_k \minmax(\Delta_k(e))\right),$$

with fixed $\beta = (1.0, 0.5, 1.0, 0.3, 0.5, 0.5)$ for (task, community-boundary proxy, connectivity, degree-based spectral proxy, representation, group), identical across datasets and not development-tuned. Zero range ($h-l < 10^{-12}$) maps to zeros (`normalize_scores`). The task-only teacher uses $y = \minmax(\Delta_{\text{task}})$.

7 S7 Teacher-edge sampling

Teacher edges are sampled without replacement by `stratified_edge_sample(src/counterfactual/sampling.py)`. Procedure for sample size n on m undirected edges:

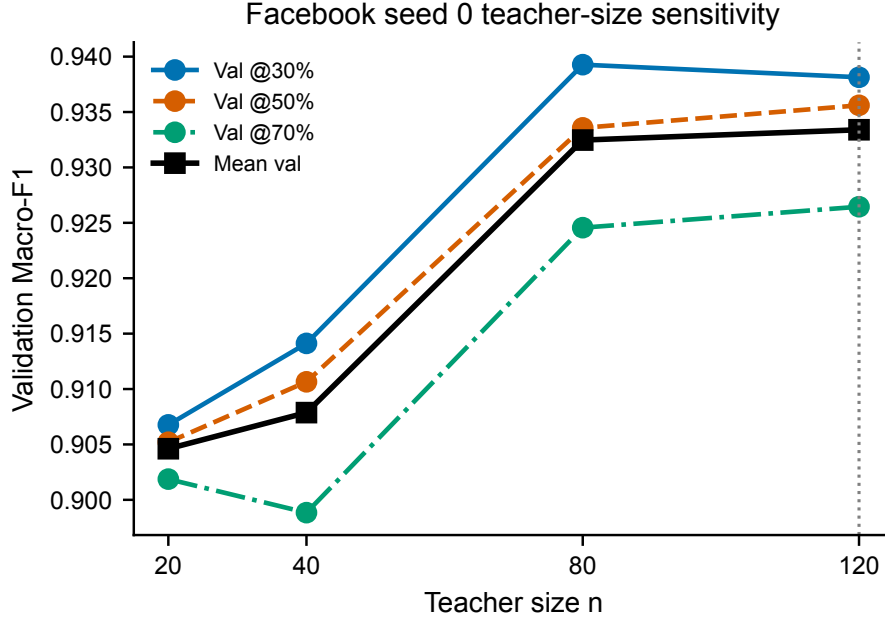
1. If $m \leq n$, return all edges.
2. Compute per-edge degree-sum $\deg(u) + \deg(v)$ and feature cosine similarity between endpoints.
3. Form 4×4 quantile strata (degree-sum quartiles $\times$ cosine quartiles; 16 nominal keys).
4. Allocate $\lfloor n / \# \text{nonempty strata} \rfloor$ (at least one when possible) by shuffling stratum keys and sampling without replacement within each stratum.
5. If fewer than n edges are selected, fill the remainder by uniform sampling without replacement from unselected edges.

Empty strata contribute nothing and are skipped. Labels, communities, and bridge status are *not* used for stratification. Reported defaults therefore implement degree $\times$ feature stratified sampling, not uniform sampling and not label- or community-stratified sampling.

8 S8 Teacher-size sensitivity

Facebook seed 0 diagnostic for $n \in \{20, 40, 80, 120\}$ at removal rates 0.3, 0.5, and 0.7. **Selection rule (as implemented):** teacher size $n=120$ was selected by ranking variants on development Macro-F1 at 50% edge removal (`scripts/run_facebook_diagnostic.py`). Mean development Macro-F1 across $\{0.3, 0.5, 0.7\}$ is reported for transparency; $n=80$ and $n=120$ are close (means 0.9325 vs 0.9334), and $n=120$ does not dominate at every removal rate. The $n=40$ result is retained as a negative pilot.

Teacher n	Dev @0.3	Test @0.3	Dev @0.5	Test @0.5	Dev @0.7	Test @0.7
20	0.907	0.901	0.905	0.900	0.902	0.897
40 (neg. pilot)	0.914	0.910	0.911	0.906	0.899	0.897
80	0.939	0.942	0.934	0.939	0.925	0.922
120 (dev.-selected)	0.938	0.941	0.936	0.938	0.926	0.922



9 S9 Edge scorer and training losses

GCN (2 layers, hidden 64, dropout 0.4); concatenation fusion in the core comparison; heteroscedastic decoder; $\mathcal{L} = 0.5 \mathcal{L}_{\text{CE}} + 1.0 \mathcal{L}_{\text{NLL}} + 0.2 \mathcal{L}_{\text{hinge}}$ with ranking margin 0.1 and 256 sampled pairs. Gated and cross-attention fusions are ablations only.

10 S10 Uncertainty estimation

$\sigma_e = \exp(\frac{1}{2} \log \sigma_e^2)$ via Gaussian NLL with variance floor 10^{-6} and $\log \sigma^2$ clamped to $[-8, 4]$. Pruning does **not** use σ_e ; ranking uses μ_e only.

11 S11 Exact-budget pruning

Algorithm 1 Unconstrained exact-budget prune (reported comparison)

Require: importance μ , removal rate r , undirected edge count m

- 1: $n_{\text{keep}} \leftarrow m - \text{round}(r \cdot m)$
 - 2: Keep the top n_{keep} edges by descending μ
-

Implementation uses `torch.argsort(importance, descending=True)` without `stable=True`. Tie order among equal μ values is therefore implementation-dependent and is not claimed to be stable. Constraint-aware mode exists in code but is disabled in the reported multi-seed comparison.

12 S12 Baseline implementation details

All baselines are converted to the same exact-budget prune: keep the top $(1 - r)|E|$ undirected edges by descending score via `budget_prune_unconstrained`. They are simplified reimplementations adapted to this protocol, not bit-for-bit reproductions of the original publications.

Random. Uniform random scores in $[0, 1]$ with seed-controlled RNG; higher score kept. No training.

ILP-GCN. Two-layer GCN encoder (hidden 64) with MLP link scorer trained by positive/negative edge BCE for `ilp_epochs` (LastFM/Facebook/GitHub: 25/20/15). Importance follows the parent ILP form $W_{\text{ILP}} = \alpha/(S+\epsilon)$ mixed with uniform initial weights ($\gamma=0.5$); higher W is retained. Code: `src/sparsification/original_ilp.py`.

NeuralSparse. Simplified task-driven scorer: one GCN layer + MLP edge head; joint Adam training with node-classification CE on the train mask minus $0.01 \times \text{mean}(\text{score})$ for `ilp_epochs` epochs (same schedule as ILP-GCN in the grid). After training, sigmoid edge scores are pruned under the exact budget. This is not the full NeuralSparse k -neighbor sampler from Zheng et al.; it is a matched-budget adaptation. Code: `src/sparsification/baselines/neuralsparse.py`.

PTDNet. Simplified topological-denoising scorer: two GCN layers + MLP denoise head; loss = train-mask CE + $0.1 \times \text{mean}(\text{score})$ for `ilp_epochs` epochs. Sigmoid scores are pruned under the exact budget. Not a full reproduction of Luo et al.’s PTDNet; matched-budget adaptation. Code: `src/sparsification/baselines/ptdnet.py`.

Resistance proxy. Effective-resistance-inspired ranking (degree-based / structural proxy path in `effective_resistance`). It is not an independent reproduction of DSpar. Duplicate records from the same resistance-based implementation were removed before aggregation ($n=10$ seeds).

13 S13 Hyperparameters and training stages

Four distinct training stages appear in the CILP pipeline (plus downstream evaluation):

Stage	Epochs (LF/FB/GH)	Role
Teacher encoder pre-train	25/25/15	Frozen encoder/classifier for counterfactual probes
Surrogate MLP fit	60/60/60	Fit <code>SurrogateEdgeImportance</code> on sampled teacher labels; expands targets to all edges for NLL; also yields S19 metrics
CILP scorer train	25/25/15	End-to-end <code>CILP</code> Social; pruning uses its μ_e
Downstream GCN eval	50/40/35	Fresh GCN on G_r after pruning
ILP/NS/PTDNet train	25/20/15	Baseline scorer epochs (not used by CILP)

“Surrogate epochs” and “scorer epochs” are therefore different models: the 60-epoch surrogate is not the ranking scorer.

Table 1: Hyperparameters used in the reported multi-seed comparison.

Group	Parameter	Value
Model	<code>encoder_type</code>	<code>gcn</code>
Model	<code>num_layers</code>	2
Model	<code>hidden_dim</code>	64
Model	<code>dropout (CILP)</code>	0.4
Model	<code>fusion (core)</code>	<code>concat</code>
Model	<code>decoder</code>	<code>ImportanceDecoder</code> hidden 64, dropout 0.2, ReLU
Model	<code>mu activation</code>	<code>sigmoid</code>
Model	<code>logvar clamp</code>	<code>[-8, 4]</code>
Model	<code>LayerNorm in fusion</code>	<code>yes (ConcatFusion)</code>
Model	<code>NodeEncoder residual/norm defaults</code>	<code>residual=True, norm=True</code>

Group	Parameter	Value
Optimization	optimizer (CILP)	Adam
Optimization	learning_rate (CILP)	1e-3
Optimization	weight_decay (CILP)	Not explicitly configured; library default used (0.0)
Optimization	scheduler	Not explicitly configured; none used
Optimization	teacher_encoder_lr	1e-2
Optimization	surrogate_mlp_epochs	60 (SurrogateEdgeImportance; not the CILP ranking scorer)
Optimization	surrogate_hidden	64
Optimization	early_stopping_patience	Not explicitly configured; fixed epoch loops
Optimization	batch_size	Not explicitly configured; full-batch graphs
Optimization	gradient_clipping	Not explicitly configured; none used
Optimization	downstream_classifier	2-layer GCN, hidden 64, dropout 0.5
Optimization	downstream_lr	1e-2
Optimization	downstream_weight_decay	5e-4
Loss	lambda_CE	0.5
Loss	lambda_NLL	1.0
Loss	lambda_rank	0.2
Loss	ranking_margin	0.1
Loss	ranking_num_pairs	256
Loss	NLL variance floor	1e-6
Teacher	teacher_n LastFM	50
Teacher	teacher_n Facebook	120
Teacher	teacher_n GitHub	40
Teacher	teacher_epochs (encoder pretrain) LF/FB/GH	25/25/15
Teacher	surrogate_mlp_epochs (all datasets)	60
Teacher	cailp_scorer_epochs LF/FB/GH	25/25/15
Teacher	downstream_epochs LastFM/Facebook/GitHub	50/40/35
Teacher	baseline_scorer_epochs (ILP/NS/PTD) LF/FB/GH	25/20/15
Teacher	sampling	stratified_edge_sample (degree $\times$ cosine quartiles) (1.0,0.5,1.0,0.3,0.5,0.5)
Teacher	beta	No; fixed across datasets
Teacher	beta tuned?	No; fixed across datasets
Teacher	zero-range handling	hi-lo; 1e-12 $\rightarrow$ zeros
Pruning	rule	keep top (1-r)—E— by descending mu
Pruning	rounding	n_keep = m - int(round(r*m))
Pruning	ties	torch.argsort without stable=True; order not guaranteed
Pruning	constraints	Disabled in reported multi-seed comparison
Statistics	CI	mean $\pm$ t_0.975,n-1 $\times$ SEM (paired t sampling dist.)
Statistics	hypothesis tests	two-sided Wilcoxon on paired diffs (primary)
Statistics	Holm family (i)	Within dataset: 4 CILP vs Task-only metrics
Statistics	Holm family (ii)	Within dataset: CILP vs baseline AUC contrasts
Statistics	effect size	dz = mean(diff)/sd(diff)

14 S14 Complete per-budget results

Dataset	Method	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
LastFM	CILP	0.792±0.019	0.792±0.024	0.786±0.040	0.761±0.052	0.769±0.061	0.743±0.070	0.715±0.071	0.671±0.067
	Task-only	0.777±0.031	0.765±0.040	0.758±0.062	0.736±0.071	0.730±0.083	0.704±0.080	0.673±0.081	0.623±0.077
	ILP-GCN	0.796±0.022	0.783±0.057	0.801±0.019	0.792±0.016	0.776±0.023	0.725±0.041	0.702±0.032	0.637±0.030
	PTDNet	0.791±0.024	0.788±0.025	0.795±0.025	0.789±0.019	0.788±0.013	0.776±0.017	0.760±0.018	0.733±0.019
	NeuralSparse	0.754±0.018	0.708±0.026	0.665±0.022	0.637±0.013	0.602±0.013	0.563±0.025	0.551±0.021	0.534±0.019
	Random	0.797±0.029	0.782±0.028	0.773±0.016	0.759±0.024	0.745±0.036	0.702±0.035	0.696±0.036	0.653±0.026
	Resistance proxy	0.741±0.024	0.658±0.037	0.613±0.031	0.590±0.027	0.558±0.027	0.530±0.026	0.524±0.020	0.497±0.014
Facebook	CILP	0.932±0.011	0.931±0.012	0.931±0.012	0.929±0.012	0.927±0.013	0.924±0.014	0.919±0.014	0.912±0.014
	Task-only	0.919±0.016	0.914±0.019	0.909±0.022	0.907±0.022	0.905±0.022	0.903±0.020	0.902±0.018	0.900±0.015
	ILP-GCN	0.939±0.004	0.939±0.004	0.938±0.004	0.936±0.004	0.933±0.002	0.927±0.005	0.920±0.006	0.909±0.003
	PTDNet	0.939±0.005	0.936±0.005	0.931±0.005	0.926±0.007	0.921±0.007	0.915±0.007	0.911±0.006	0.906±0.007
	NeuralSparse	0.929±0.005	0.924±0.004	0.917±0.005	0.906±0.003	0.898±0.005	0.891±0.006	0.888±0.006	0.884±0.005
	Random	0.935±0.005	0.932±0.005	0.928±0.005	0.922±0.006	0.917±0.006	0.911±0.006	0.903±0.006	0.897±0.005
	Resistance proxy	0.899±0.005	0.889±0.005	0.888±0.006	0.888±0.006	0.886±0.007	0.886±0.006	0.888±0.007	0.889±0.007
GitHub	CILP	0.814±0.006	0.813±0.006	0.812±0.005	0.811±0.005	0.809±0.006	0.809±0.006	0.809±0.005	0.806±0.004
	Task-only	0.811±0.007	0.809±0.006	0.807±0.008	0.808±0.007	0.806±0.007	0.806±0.006	0.804±0.005	0.803±0.006
	ILP-GCN	0.816±0.007	0.818±0.005	0.816±0.005	0.815±0.004	0.813±0.005	0.812±0.006	0.810±0.005	0.808±0.003
	PTDNet	0.816±0.005	0.818±0.004	0.817±0.005	0.815±0.006	0.813±0.004	0.812±0.005	0.809±0.004	0.807±0.003
	NeuralSparse	0.812±0.006	0.809±0.006	0.806±0.005	0.805±0.007	0.802±0.005	0.796±0.006	0.794±0.006	0.785±0.008
	Random	0.814±0.004	0.813±0.005	0.811±0.006	0.810±0.006	0.807±0.005	0.807±0.004	0.804±0.005	0.806±0.004
	Resistance proxy	0.806±0.007	0.806±0.006	0.806±0.005	0.808±0.005	0.807±0.005	0.808±0.004	0.808±0.004	0.808±0.005

15 S15 Full per-seed results

Machine-readable: `results/processed/authoritative_results.csv` (1,890 rows; unique key `dataset×method×seed×edge_removal_rate`). The Stage A freeze archive and checksums are recorded in `docs/stage_a_freeze`.

16 S16–S18 Paired statistical comparisons

Hypothesis tests: two-sided Wilcoxon signed-rank on paired seed-wise differences (SciPy `wilcoxon`; `zero_method='wilcoxon'`, `correction=False`, `method='auto'`). Confidence intervals: mean difference $\pm t_{0.975,n-1}$ SEM from the paired t sampling distribution; paired t p -values are computed in the analysis scripts but are not the primary reported tests. Holm family (ii) adjusts Wilcoxon p -values across CILP-versus-baseline AUC contrasts within each dataset.

Table 2: Paired contrasts (CILP minus comparator) on common seeds.

Dataset	Contrast	n	Δ	95% CI	Wilcoxon p	Holm p	d_z
facebook	AUC vs Task-only	10	+0.0142	[+0.0053,+0.0231]	0.00390625	0.015625	1.137
facebook	AUC vs ILP-GCN	10	-0.0029	[-0.0091,+0.0033]	0.625	1	-0.333
facebook	AUC vs PTDNet	10	+0.0026	[-0.0052,+0.0104]	0.625	1	0.235
facebook	AUC vs NeuralSparse	10	+0.0177	[+0.0113,+0.0241]	0.00195312	0.011719	1.987
facebook	AUC vs Random	10	+0.0069	[+0.0006,+0.0131]	0.0488281	0.14648	0.783
facebook	AUC vs Resistance proxy	10	+0.0283	[+0.0229,+0.0336]	0.00195312	0.011719	3.786
github	AUC vs Task-only	10	+0.0033	[+0.0015,+0.0051]	0.00195312	0.011719	1.292
github	AUC vs ILP-GCN	10	-0.0024	[-0.0040,-0.0008]	0.00390625	0.015625	-1.104
github	AUC vs PTDNet	10	-0.0023	[-0.0039,-0.0007]	0.00585938	0.017578	-1.029
github	AUC vs NeuralSparse	10	+0.0084	[+0.0062,+0.0106]	0.00195312	0.011719	2.707
github	AUC vs Random	10	+0.0010	[+0.0001,+0.0019]	0.105469	0.10547	0.800
github	AUC vs Resistance proxy	10	+0.0022	[+0.0002,+0.0042]	0.0371094	0.074219	0.805
lastfm	AUC vs Task-only	10	+0.0265	[+0.0022,+0.0508]	0.00390625	0.015625	0.781
lastfm	AUC vs ILP-GCN	10	+0.0029	[-0.0292,+0.0350]	0.431641	0.55078	0.065
lastfm	AUC vs PTDNet	10	-0.0232	[-0.0527,+0.0063]	0.0644531	0.19336	-0.563
lastfm	AUC vs NeuralSparse	10	+0.1037	[+0.0748,+0.1326]	0.00195312	0.011719	2.567
lastfm	AUC vs Random	10	+0.0128	[-0.0160,+0.0416]	0.275391	0.55078	0.318
lastfm	AUC vs Resistance proxy	10	+0.1344	[+0.1041,+0.1648]	0.00195312	0.011719	3.171

17 S19 Surrogate-quality metrics

Dataset	Method	n	MAE	Spearman	Kendall	ECE
lastfm	CILP	10	0.076	0.775	0.632	0.077
lastfm	Task-only	10	0.115	0.318	0.225	0.147
facebook	CILP	10	0.106	0.651	0.525	0.097
facebook	Task-only	10	0.164	0.104	0.068	0.175
github	CILP	10	0.237	0.255	0.211	0.165
github	Task-only	10	0.217	0.046	0.036	0.165

On GitHub, CILP has higher Spearman/Kendall agreement while Task-only has lower raw MAE and ECE is essentially tied. Because sparsification ranks edges for deletion, rank correlation is the primary surrogate diagnostic; absolute MAE can favour flat predictors that are poor rankers. Top- k overlap and normalized MAE relative to target scale are recommended extensions.

18 S20 Ablation study

LastFM ablations (seeds 0–4) are grouped scientifically below. Facebook ablations were incomplete at the time of analysis. Adversarial variants were omitted after a negative pilot.

Group	Scientific name	Teacher	Prior	Fusion	Seeds	Status
Fusion	Concatenation	multi-criteria	none	concat	0–4	complete
Fusion	Gated fusion	multi-criteria	none	gated	0–4	complete
Fusion	Cross-attention	multi-criteria	none	cross-attn	0–4	complete
Node prior	No node prior	multi-criteria	none	concat	0–4	complete
Node prior	Analytical mass	multi-criteria	analytical	concat	0–4	complete
Node prior	Degree prior	multi-criteria	degree	concat	0–4	complete
Node prior	PageRank prior	multi-criteria	PageRank	concat	0–4	complete
Node prior	Black Hole-inspired	multi-criteria	Black Hole	concat	0–4	complete
Teacher	Multi-criteria	multi-criteria	none	concat	0–4	complete
Teacher	Task-only	task-only	none	concat	0–4	complete
Optional	Adversarial variants	—	—	—	—	rejected / skipped
Context	ILP-GCN	N/A	N/A	N/A	0–4	complete

Budgets reported in ablation figures: 0.3, 0.5, 0.7. Repository identifiers for these variants appear only in S29.

19 S21 Multi-criteria versus task-only supervision

We tested whether combining multiple deletion-importance criteria improves edge ranking relative to task-only supervision under the same encoder, fusion, optimization, budgets, and paired seeds. **Family (i) definition:** for each dataset, four paired metrics (CILP – Task-only) form one family: (i) sparsity–Macro-F1 AUC; (ii) giant-component ratio at 50%; (iii) bridge retention at 50%; (iv) minority-degree retention at 50%. Holm adjustment is applied to Wilcoxon p -values within each dataset family (i). **Family (ii):** within each dataset, CILP-versus-baseline AUC contrasts form a separate Holm family (Supplementary Information S16–S18); adjusted p -values are not interchangeable with family (i). **Decision rule:** multi-criteria supervision is supported across datasets when at least two metrics are supported (positive Δ , Holm $p < 0.05$) on at least two datasets. The analysis rule was fixed before completion and integration of the GitHub grid; we do not claim timestamped preregistration beyond the Stage A freeze record. On LastFM, Facebook, and GitHub the within-dataset criterion is met (4/4 metrics each); multi-criteria supervision therefore satisfies the fixed support criterion on all three datasets. This support is relative to the matched Task-only teacher and does not imply that CILP outperforms all established sparsifiers.

Table 3: Paired CILP versus Task-only tests (common seeds). Family: four metrics per dataset; Holm adjustment on Wilcoxon p .

Dataset	Metric	CILP	Task-only	Δ	95% CI	Wilcoxon p	Holm p	d_z	W/T/L
lastfm	AUC	0.593	0.566	+0.0265	[+0.0022,+0.0508]	0.003906	0.007812	0.781	9/0/1
lastfm	GC @50%	0.928	0.772	+0.1564	[+0.0165,+0.2962]	0.001953	0.007812	0.800	10/0/0
lastfm	Bridge @50%	0.904	0.681	+0.2230	[+0.0318,+0.4142]	0.001953	0.007812	0.835	10/0/0
lastfm	Min-deg @50%	0.891	0.673	+0.2182	[+0.0149,+0.4214]	0.01151	0.01151	0.768	8/2/0
facebook	AUC	0.739	0.725	+0.0142	[+0.0053,+0.0231]	0.003906	0.01562	1.137	9/0/1
facebook	GC @50%	0.923	0.626	+0.2971	[+0.1321,+0.4622]	0.007686	0.02306	1.288	9/1/0
facebook	Bridge @50%	0.827	0.422	+0.4053	[+0.1938,+0.6169]	0.007686	0.02306	1.371	9/1/0
facebook	Min-deg @50%	0.650	0.493	+0.1569	[+0.0204,+0.2935]	0.01367	0.02306	0.822	8/0/2
github	AUC	0.648	0.645	+0.0033	[+0.0015,+0.0051]	0.001953	0.007812	1.292	10/0/0
github	GC @50%	0.903	0.674	+0.2284	[+0.0717,+0.3851]	0.001953	0.007812	1.043	10/0/0
github	Bridge @50%	0.802	0.483	+0.3195	[+0.1305,+0.5085]	0.001953	0.007812	1.209	10/0/0
github	Min-deg @50%	0.565	0.452	+0.1136	[+0.0270,+0.2002]	0.003906	0.007812	0.938	9/0/1

20 S22 Structural and group metrics

Means at 50% removal appear below; multi-budget (30/50/70%) giant-component and bridge values are also reported in the main manuscript.

Dataset	Method	Giant-component ratio	Bridge retention	Minority-degree retention
lastfm	CILP	0.928	0.904	0.891
lastfm	Task-only	0.772	0.681	0.673
lastfm	ILP-GCN	0.880	0.920	0.909
lastfm	PTDNet	0.822	0.759	0.914
lastfm	Random	0.784	0.501	0.500
facebook	CILP	0.923	0.827	0.650
facebook	Task-only	0.626	0.422	0.493
facebook	ILP-GCN	0.966	0.979	0.964
facebook	PTDNet	0.754	0.607	0.630
facebook	Random	0.863	0.502	0.500
github	CILP	0.903	0.802	0.565
github	Task-only	0.674	0.483	0.452
github	ILP-GCN	0.925	0.685	0.666
github	PTDNet	0.925	0.687	0.746
github	Random	0.873	0.500	0.499

21 S23 Runtime

`train_seconds` includes teacher construction and scorer fitting for CILP and Task-only; these components were not separately logged, so dominance cannot be attributed to teacher construction alone.

Dataset	Method	n	Train seconds	Prune seconds
lastfm	Random	10	0.01±0.01	0.193±0.138
lastfm	ILP-GCN	10	5.20±2.05	0.052±0.088
lastfm	PTDNet	10	2.57±0.72	0.190±0.174
lastfm	NeuralSparse	10	4.09±1.41	0.053±0.097
lastfm	Task-only	10	254.63±149.00	0.115±0.140
lastfm	CILP	10	198.40±89.25	0.158±0.140
facebook	Random	10	0.00±0.00	0.107±0.118
facebook	ILP-GCN	10	8.55±1.95	0.068±0.103
facebook	PTDNet	10	5.09±1.44	0.206±0.299
facebook	NeuralSparse	10	6.56±1.95	0.109±0.142
facebook	Task-only	10	1635.33±541.98	0.030±0.021
facebook	CILP	10	1436.51±421.38	0.050±0.073
github	Random	10	0.00±0.01	0.037±0.005
github	ILP-GCN	10	7.43±0.34	0.034±0.004
github	PTDNet	10	3.87±0.29	0.036±0.006
github	NeuralSparse	10	4.82±0.29	0.036±0.005
github	Task-only	10	504.15±93.94	0.034±0.003
github	CILP	10	503.97±85.02	0.034±0.002

22 S24 Failure cases and rejected hypotheses

Adversarial pilot (LastFM seed 0, 50%): Macro-F1 ≈ 0.666 vs core ≈ 0.829 . Facebook teacher $n=40$ underperformed the development-selected $n=120$ setting (S8). Stage A GitHub grid: no method failures across seeds 0–9, seven methods, and budgets 0.1–0.9.

23 S25 Constraint shortfalls

Reported comparison: constraints off; achieved removals match requested budgets exactly under unconstrained ranking.

24 S26 Reproducibility checklist

Processed authoritative CSV/JSON (1,890 rows); Stage A freeze manifest with SHA-256 checksums; consistency validator; fixed splits for all three datasets; negative pilots retained; figure provenance notes. Paths: `docs/stage_a_freeze_manifest.md`, `docs/figure_data_provenance.md`.

25 S27 Software and hardware

Reported multi-seed runs used CPU execution (`device=cpu` in experiment logs); MPS is disabled in the device helper because several PyG scatter operations are unsupported. Exact CPU model, core counts, and RAM for the machines that produced every grid row are not stored as structured metadata in the repository and therefore cannot be uniquely recovered. Environment specification files declare Python 3.10 (`environment.yml`, `Dockerfile`); analysis regeneration on the present workstation used Python 3.9.6 with: PyTorch 2.4.0, PyTorch Geometric 2.6.1, NetworkX 3.2.1, NumPy 1.26.4, SciPy 1.13.1, scikit-learn 1.6.1, matplotlib 3.9.4. No `pip freeze` lockfile is present; `requirements.txt` lists version floors only. BLAS backend was not recorded in experiment logs.

26 S28 Data and code availability details

Public SNAP/MUSAE releases for LastFM, Facebook, and GitHub. Processed graphs: `data/processed/`. Fixed splits: `data/splits/`. Authoritative store: `results/processed/authoritative_results.csv` (1,890 rows). GitHub raw per-seed outputs: `results/raw/grid/`. Stage A freeze archive: `results/frozen/`. Regenerated figures/tables: `paper/`. Regeneration and consistency-check scripts: `scripts/`. No DOI is claimed in this draft.

27 S29 Repository ID mapping

Internal experiment identifiers appear only in this table.

Table 4: Repository ID mapping (internal identifiers; not used elsewhere in the PDFs).

Repository ID	Scientific description	Notes
A1_full_core	CILP core configuration	Reported main method
A2_original_ilp	ILP-GCN parent baseline	Parent method
A8_concat	Multi-criteria teacher with concatenation	Fusion-matched to task-only
A9_gated	Multi-criteria teacher with gated fusion	Ablation
A10_cross_attn	Multi-criteria teacher with cross-attention	Ablation; not core
A19_no_mass	No node prior	Ablation
A20_analytical_mass	Analytical mass prior	Ablation
A23_degree_prior	Degree prior	Ablation

Repository ID	Scientific description	Notes
A24_pagerank_prior	PageRank prior	Ablation
A25_no_adversarial	No adversarial module	Equivalent to core path
A26_repr_disc	Representation discriminator	Skipped; rejected pilot
A27_mask_gan	Mask GAN	Skipped; rejected pilot
A28_gan_without_ilp	GAN without ILP	Skipped; rejected pilot
A31_single_obj_concat	Task-only counterfactual teacher	Matched ablation
A31_single_obj_gated	Task-only with gated fusion	Ablation
A32_black_hole_mass	Black Hole-inspired prior	Ablation
cailp_multi	CILP main grid key	JSON method key
cailp_a31	Task-only grid key	JSON method key

References

Benedek Rozemberczki, Carl Allen, and Rik Sarkar. Multi-scale attributed node embedding. *Journal of Complex Networks*, 9(2):cnab014, 2021. doi: 10.1093/comnet/cnab014.